

ICPC 2026					
PROBLEM	READ	SOLVED	WRITTEN	OPENED	COMMENT
A					
B					
C					
D					
E					
F					
G					
H					
I					
J					
K					
L					
M					
N					

Time	Andrey	Taisia	Timofey
0:30			
1:00			
1:30			
2:00			
2:30			
3:00			
3:30			
4:00			
4:30			
5:00			

Contents	
1	Algebra 2
1.1	Gauss 2
2	Combinatorics 2
2.1	Gray Code 2
3	Convex Hull Trick 2
3.1	Convex Hull Trick 2
4	Convolutions 3
4.1	FFT MOD 3
4.2	FFT 4
4.3	FHT 5
4.4	SOS DP 6
5	Data Structures 6
5.1	Centroid Decomposition 6
5.2	DSU 7
5.3	Disjoint Sparse Table 7
5.4	Fenwick 2D 8
5.5	Fenwick Tree 9
5.6	HLD 9
5.7	Implicit Segment Tree 10
5.8	Persistent DSU 12
5.9	Prefix Automate 12
5.10	Segment Tree Beats 13
5.11	Segment Tree Mass 15
5.12	Segment Tree 17
5.13	Sparse Table 18
5.14	Treap 18
6	Flows 20
6.1	Dinic 20
7	Graphs 21
7.1	2 SAT 21
7.2	Euler Path 22
7.3	Hopcroft Karp 22
7.4	Hungarian 23
7.5	Initial Pairing 23
7.6	Kuhn 24
8	Number Theory 25

8.1	Eratosthenes	25
8.2	FloorSum	25
8.3	Miller Rabin	25
9	Strings	26
9.1	LCP	26
9.2	Prefix Function	26
9.3	Suffix Massive	26
9.4	Z Function	27

1 Algebra

1.1 Gauss

```
// a: n x (m + 1), last column is free terms
// returns one solution (free vars = 0), empty vector if no solutions
vector<ld> gauss(vector<vector<ld>> a) {
    const ld EPS = 1e-12;
    ll n = a.size();
    if (n == 0) {
        return {};
    }
    ll m = a[0].size() - 1;

    vector<ll> where(m, -1);
    for (ll col = 0, row = 0; col < m && row < n; ++col) {
        ll sel = row;
        for (ll i = row; i < n; ++i) {
            if (fabs1(a[i][col]) > fabs1(a[sel][col]))
                sel = i;
        }
        if (fabs1(a[sel][col]) < EPS)
            continue;

        swap(a[sel], a[row]);
        where[col] = row;

        for (ll i = 0; i < n; ++i) {
            if (i == row)
                continue;
            ld c = a[i][col] / a[row][col];
```

```
            if (fabs1(c) < EPS)
                continue;
            for (ll j = col; j <= m; ++j)
                a[i][j] -= a[row][j] * c;
        }
        ++row;
    }

    vector<ld> ans(m, 0);
    for (ll i = 0; i < m; ++i) {
        if (where[i] != -1)
            ans[i] = a[where[i]][m] / a[where[i]][i];
    }

    for (ll i = 0; i < n; ++i) {
        ld sum = 0;
        for (ll j = 0; j < m; ++j)
            sum += ans[j] * a[i][j];
        if (fabs1(sum - a[i][m]) > EPS)
            return {};
    }
    return ans;
}
```

2 Combinatorics

2.1 Gray Code

```
inline ll g(ll n) {
    return n ^ (n >> 1);
}
```

3 Convex Hull Trick

3.1 Convex Hull Trick

```
// monotone lines
struct ConvexHullTrick {
    // a, b of length k, p of lengths max(0, k - 1)
    // all lines is a[i] * x + b[i]
    // and line i > 0 maximum from x = p[i - 1] to x = p[i] - 1
```

```

vector<ll> a, b, p;

// a1 < a2
static ll intersect(ll a1, ll b1, ll a2, ll b2) {
    ll den = a2 - a1;
    ll num = b1 - b2;
    if (num >= 0) {
        return (num + den - 1) / den;
    } else {
        return num / den;
    }
}

// na >= a[i] forall i
void add_line(ll na, ll nb) {
    if (a.empty()) {
        a.push_back(na);
        b.push_back(nb);
        return;
    }
    if (a.back() > na) {
        return;
    }
    while (!p.empty() && a.back() * p.back() + b.back() <= na * p.
back() + nb) {
        p.pop_back();
        a.pop_back();
        b.pop_back();
    }
    // else na == a.back() and in one concrete point
    // new line leq of old > in all points new line >
    if (a.back() < na) {
        p.push_back(intersect(a.back(), b.back(), na, nb));
        a.push_back(na);
        b.push_back(nb);
    }
}

ll query(ll x) {
    if (a.empty()) {
        return -inf;
    }
    ll l = 0, r = (ll)a.size();
    while (r - l > 1) {

```

```

        ll m = (r + l) / 2;
        if (p[m - 1] <= x) {
            l = m;
        } else {
            r = m;
        }
    }
    return a[l] * x + b[l];
};

```

4 Convolutions

4.1 FFT MOD

```

const ll MOD = 998244353;
const ll ROOT = 31;
const int ROOT_PW = (1ll << 23);

vector<vector<ll>> w;

void precalc_ws(ll lg_n) {
    w.resize(lg_n + 1);
    w[0].resize(1, 1);
    for (int l = 1; l <= lg_n; ++l) {
        ll n = (1ll << l);
        w[l].resize(n / 2);

        ll fw = ROOT;
        for (int i = 0; i < 23 - l; ++i) {
            fw = (fw * fw) % MOD;
        }
        for (int j = 0; j < (n >> 1); ++j) {
            if (j % 2 == 0) {
                w[l][j] = w[l - 1][j / 2];
            } else {
                w[l][j] = (w[l - 1][j / 2] * fw) % MOD;
            }
        }
    }
}

void fft_inpl(vector<ll>& a, bool invert) {

```

```

int n = a.size();

// apply permutation by bit_rev
for (int i = 1, j = 0; i < n; ++i) {
    int bit = n >> 1;
    for (; j & bit; bit >>= 1)
        j ^= bit;
    j ^= bit;
    if (i < j)
        swap(a[i], a[j]);
}

for (int len = 2, lg = 1; len <= n; len <= 1, ++lg) {
    for (int i = 0; i < n; i += len) {
        for (int j = 0; j < (len >> 1); ++j) {
            ll root = w[lg][j];
            if (invert && j) {
                root = MOD - w[lg][(len >> 1) - j];
            }
            ll u = a[i + j], v = (a[i + j + len / 2] * root) % MOD;
            a[i + j] = u + v < MOD ? u + v : u + v - MOD;
            a[i + j + len / 2] = u - v >= 0 ? u - v : u - v + MOD;
        }
    }
}

if (invert) {
    int nrev = rev(n);
    for (int i = 0; i < n; ++i) {
        a[i] *= nrev;
        a[i] %= MOD;
    }
}

vector<ll> vect_mult(const vector<ll>& a, const vector<ll>& b) {
    ll sz = a.size() + b.size() - 1;
    ll n = 1;
    int lg_n = 0;
    while (n < sz) {
        n <= 1;
        ++lg_n;
    }
    if (w.size() <= lg_n) {
        precalc_ws(lg_n);

```

```

    }

    vector<ll> fa(n, 0), fb(n, 0);
    for (int i = 0; i < n; ++i) {
        if (i < a.size()) {
            fa[i] = a[i];
        }
        if (i < b.size()) {
            fb[i] = b[i];
        }
    }

    fft_inpl(fa, false);
    fft_inpl(fb, false);
    for (int i = 0; i < n; ++i) {
        fa[i] *= fb[i];
        fa[i] %= MOD;
    }
    fft_inpl(fa, true);

    vector<ll> mult(sz);
    for (int i = 0; i < sz; ++i) {
        mult[i] = fa[i];
    }
    return mult;
}

```

4.2 FFT

```

vector<vector<comp>> w;

void precalc_ws(ll lg_n) {
    w.resize(lg_n + 1);
    w[0].resize(1, 1);
    for (int l = 1; l <= lg_n; ++l) {
        ll n = (1ll << l);
        w[l].resize(n / 2);

        // need cosl, sinl for complex<ld>
        comp fw(cos(2 * PI / n), sin(2 * PI / n));
        for (int j = 0; j < (n >> 1); ++j) {
            if (j % 2 == 0) {
                w[l][j] = w[l - 1][j / 2];
            } else {

```

```

        w[l][j] = w[l - 1][j / 2] * fw;
    }
}

void fft_inpl(vector<comp>& a, bool invert) {
    int n = a.size();

    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;
        if (i < j)
            swap(a[i], a[j]);
    }

    for (int len = 2, lg = 1; len <= n; len <<= 1, ++lg) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < (len >> 1); ++j) {
                comp root = w[lg][j];
                if (invert && j)
                    root = -w[lg][(len >> 1) - j];
                comp u = a[i + j], v = a[i + j + (len >> 1)] * root;
                a[i + j] = u + v;
                a[i + j + (len >> 1)] = u - v;
            }
        }
    }

    if (invert) {
        double rev_n = 1.0 / n;
        for (int i = 0; i < n; ++i) {
            a[i] *= rev_n;
        }
    }
}

vector<ll> vect_mult(const vector<ll>& a, const vector<ll>& b) {
    ll sz = a.size() + b.size() - 1;
    ll n = 1;
    int lg_n = 0;
    while (n < sz) {
        n <<= 1;

```

```

        ++lg_n;
    }
    if (w.size() <= lg_n) {
        precalc_ws(lg_n);
    }

    vector<comp> p(n, 0);
    for (int i = 0; i < n; ++i) {
        p[i] = comp(i < a.size() ? a[i] : 0, i < b.size() ? b[i] : 0);
    }

    fft_inpl(p, false);
    for (int i = 0; i < n; ++i) {
        p[i] *= p[i];
    }
    fft_inpl(p, true);

    vector<ll> mult(sz);
    for (int i = 0; i < sz; ++i) {
        // need roundl for complex<ld>
        mult[i] = round(p[i].imag() * 0.5);
    }
    return mult;
}

4.3 FHT

inline ll soft_mod(ll a) {
    return a >= MOD ? a - MOD : a;
}

void FHT(vector<ll>& a) {
    ll sz = a.size();
    for (int mid = 1; mid < sz; mid <<= 1) {
        for (int i = 0; i < sz; i += mid << 1) {
            for (int j = 0; j < mid; ++j) {
                ll x = a[i + j];
                ll y = a[i + j + mid];
                a[i + j] = soft_mod(x + y);
                a[i + j + mid] = soft_mod(x - y + MOD);
            }
        }
    }
}

```

```

vector<ll> xor_mult(vector<ll> a, vector<ll> b) {
    FHT(a);
    FHT(b);
    ll n = a.size();
    vector<ll> c(n);

    for (int i = 0; i < n; ++i) {
        c[i] = (a[i] * b[i]) % MOD;
    }

    FHT(c);
    ll revn = bin_pow(n, MOD - 2);
    for (int i = 0; i < n; ++i) {
        c[i] *= revn;
        c[i] %= MOD;
    }
    return c;
}

```

4.4 SOS DP

```

void sos(vector<ll>& dp) {
    ll n = dp.size();
    ll k = 0;
    while ((1ll << k) < n) {
        ++k;
    }

    for (int j = 0; j < k; ++j) {
        for (int i = 0; i < n; ++i) {
            if (i & (1ll << j)) {
                dp[i] += dp[i ^ (1ll << j)];
            }
        }
    }
}

void revsos(vector<ll>& dp) {
    ll n = dp.size();

    ll k = 0;
    while ((1ll << k) < n) {
        ++k;
    }
}

```

```

    }

    for (int j = k - 1; j >= 0; --j) {
        for (int i = 0; i < n; ++i) {
            if (i & (1ll << j)) {
                dp[i] -= dp[i ^ (1ll << j)];
            }
        }
    }
}

```

5 Data Structures

5.1 Centroid Decomposition

```

struct CentroidDecomp {
    ll n;
    vector<ll> parent;

    void calc_sizes(ll v, ll mark, vector<ll>& used, vector<vector<ll>>&
g, vector<ll>& s) {
        s[v] = 1;
        used[v] = mark;

        for (ll u : g[v]) {
            if (used[u] < mark) {
                calc_sizes(u, mark, used, g, s);
                s[v] += s[u];
            }
        }
    }

    ll find_centroid(ll v, ll mark, vector<ll>& used, vector<vector<ll>>&
g, vector<ll>& s, ll sz) {
        used[v] = mark;

        for (ll u : g[v]) {
            if (used[u] < mark) {
                if (s[u] > sz / 2) {
                    return find_centroid(u, mark, used, g, s, sz);
                }
            }
        }
    }
}

```

5.2 DSU

```

    return v;
}

void set_parent(ll v, ll mark, vector<ll>& used, vector<vector<ll>>&
g, ll cent) {
    if (v != cent)
        parent[v] = cent;
    used[v] = mark;

    for (ll u : g[v]) {
        if (used[u] < mark) {
            set_parent(u, mark, used, g, cent);
        }
    }
}

CentroidDecomp(ll N, vector<vector<ll>>& g) {
    n = N;
    parent.assign(n, -1);

    vector<ll> used(n, 0);
    vector<ll> s(n, 0);
    ll cnt_used = 0;
    ll mark = 0;

    while (cnt_used < n) {
        ++mark;
        ll pass_mark = mark;
        for (int i = 0; i < n; ++i) {
            if (used[i] >= pass_mark)
                continue;
            calc_sizes(i, pass_mark, used, g, s);
            ++mark;
            ll cent = find_centroid(i, mark, used, g, s, s[i]);

            ++mark;
            set_parent(i, mark, used, g, cent);
            used[cent] = inf;
            ++cnt_used;
        }
    }
};

```

```

struct DSU {
    ll n;
    vector<ll> prev, size;

    DSU(ll N) : n(N) {
        prev.assign(n, -1);
        size.assign(n, 1);
    }

    ll root(ll a) {
        ll aa = a;
        while (prev[a] != -1)
            a = prev[a];

        while (prev[aa] != -1) {
            ll last = prev[aa];
            prev[aa] = a;
            aa = last;
        }
        return a;
    }

    bool connected(ll a, ll b) {
        return root(a) == root(b);
    }

    bool unite(ll a, ll b) {
        a = root(a);
        b = root(b);
        if (a == b)
            return false;
        if (size[a] < size[b])
            swap(a, b);
        prev[b] = a;
        size[a] += size[b];
        return true;
    }
};

```

5.3 Disjoint Sparse Table

```

struct DisjointSparseTable {
    vector<vector<ll>> dp;

    ll neutral() {
        return 0;
    }

    ll merge(ll a, ll b) {
        return a + b;
    }

    DisjointSparseTable() {}

    DisjointSparseTable(const vector<ll>& a) {
        ll n = a.size();
        ll k = 0;
        ll nn = 1;
        while (nn <= n) {
            ++k;
            nn <<= 1;
        }

        dp.assign(k, vector<ll>(n + 1, neutral()));

        ll t = 0;
        for (ll len = 1; len <= n; len <<= 1, ++t) {
            for (ll cent = len; cent <= n; cent += len * 2) {
                dp[t][cent - 1] = a[cent - 1];
                for (ll i = cent - 2; i >= cent - len; --i) {
                    dp[t][i] = merge(a[i], dp[t][i + 1]);
                }

                dp[t][cent] = neutral();
                for (ll i = cent + 1; i < min(cent + len, n + 1); ++i) {
                    dp[t][i] = merge(dp[t][i - 1], a[i - 1]);
                }
            }
        }

        // query on [l, r)
        ll query(ll l, ll r) {
            if (r <= 1)

```

```

                return neutral();
                ll t = 63 - __builtin_clzll((ull)(1 ^ r));
                return merge(dp[t][l], dp[t][r]);
            }
        };
    };
};

```

5.4 Fenwick 2D

```

template <class T>
struct Fenwick2D {
    ll n = 0, m = 0;
    vector<vector<T>> bit;

    Fenwick2D() = default;
    Fenwick2D(int n_, int m_) {
        init(n_, m_);
    }

    void init(int n_, int m_) {
        n = n_;
        m = m_;
        bit.assign((int)n + 1, vector<T>((int)m + 1, T{}));
    }

    // add delta at (x, y), 0-indexed
    void add(ll x, ll y, T delta) {
        for (ll i = x + 1; i <= n; i += i & -i)
            for (ll j = y + 1; j <= m; j += j & -j)
                bit[i][j] += delta;
    }

    // sum over [0..x] x [0..y], 0-indexed; returns 0 if x<0 or y<0
    T sum_prefix(ll x, ll y) const {
        if (x < 0 || y < 0)
            return T{};
        T res{};
        for (ll i = x + 1; i > 0; i -= i & -i)
            for (ll j = y + 1; j > 0; j -= j & -j)
                res += bit[i][j];
        return res;
    }

    // sum over rectangle [x1..x2] x [y1..y2], inclusive, 0-indexed
    T sum_rect(ll x1, ll y1, ll x2, ll y2) const {

```

```

        return sum_prefix(x2, y2) - sum_prefix(x1 - 1, y2) - sum_prefix(
x2, y1 - 1) + sum_prefix(x1 - 1, y1 - 1);
    }
};

```

5.5 Fenwick Tree

```

struct FenwickTree {
    vector<ll> tree;
    ll n;

    FenwickTree(ll N) {
        n = N;
        tree.assign(n, 0);
    }

    ll F(ll x) {
        return x & (x + 1);
    }

    ll H(ll x) {
        return x | (x + 1);
    }

    // [0, r)
    ll query(ll r) {
        if (r > n) {
            r = n;
        }
        --r;
        ll ans = 0;
        while (r >= 0) {
            ans ^= tree[r];
            r = F(r) - 1;
        }
        return ans;
    }

    // [l, r)
    ll query(ll l, ll r) {
        return query(r) ^ query(l);
    }

    void add(ll id, ll val) {

```

```

        while (id < n) {
            tree[id] ^= val;
            id = H(id);
        }
    }
};

```

5.6 HLD

```

struct HLD {
    ll n;
    vector<ll> top, pos, h, par;

    ll gpos = 0;

    SegmentTree st;

    void calc_sizes(ll v, vector<char>& used, const vector<vector<ll>>& g
, vector<ll>& s) {
        s[v] = 1;
        used[v] = 1;

        for (ll u : g[v]) {
            if (!used[u]) {
                calc_sizes(u, used, g, s);
                s[v] += s[u];
            }
        }
    }

    void build_dec(ll v, ll p, ll pr, vector<char>& used, const vector<
vector<ll>>& g, vector<ll>& s) {
        used[v] = 1;
        pos[v] = gpos++;
        top[pos[v]] = pos[p];
        par[pos[v]] = pos[pr];
        if (pr != v)
            h[pos[v]] = h[pos[pr]] + 1;

        vector<ll> child;

        for (ll u : g[v]) {
            if (!used[u]) {
                child.push_back(u);

```

```

    }
}

for (int i = 1; i < child.size(); ++i) {
    if (s[child[i]] > s[child[0]]) {
        swap(child[i], child[0]);
    }
}

if (!child.empty()) {
    build_dec(child[0], p, v, used, g, s);
    for (int i = 1; i < child.size(); ++i) {
        build_dec(child[i], child[i], v, used, g, s);
    }
}
}

HLD(ll N, const vector<vector<ll>>& g, const vector<ll>& a) {
    n = N;

    top.resize(n, 0);
    pos.resize(n, 0);
    h.resize(n, 0);
    par.resize(n, 0);
    vector<char> used(n, 0);
    vector<ll> s(n, 0);

    calc_sizes(0, used, g, s);
    used.assign(n, 0);
    build_dec(0, 0, 0, used, g, s);

    vector<ll> pa(n, 0);
    for (int i = 0; i < n; ++i) {
        pa[pos[i]] = a[i];
    }

    st = SegmentTree(pa);
}

ll query(ll u, ll v) {
    ll ans = 0;

    u = pos[u];
    v = pos[v];

```

```

    ll pu = top[u];
    ll pv = top[v];

    while (pu != pv) {
        if (h[pu] >= h[pv]) {
            ans += st.query(pu, u + 1).sum;
            u = par[pu];
            pu = top[u];
        } else {
            ans += st.query(pv, v + 1).sum;
            v = par[pv];
            pv = top[v];
        }
    }
    ans += st.query(min(u, v), max(u, v) + 1).sum;

    return ans;
}

void change(ll u, ll val) {
    u = pos[u];
    st.change(u, val);
}
};

```

5.7 Implicit Segment Tree

```

struct ImplicitSegmentTree {
    struct Node {
        Node* left = nullptr;
        Node* right = nullptr;
        ll sum = 0;
        ll add = 0;
    };

    Node neutral = Node();
    ll n;
    Node* root = nullptr;

    Node merge(Node n1, Node n2) {
        Node res;
        res.sum = n1.sum + n2.sum;
        return res;
    }
};

```

```

}

void fix(Node* node) {
    node->sum = 0;
    if (node->left) {
        node->sum += node->left->sum;
    }
    if (node->right) {
        node->sum += node->right->sum;
    }
}

void apply(Node* node, ll l, ll r, ll val) {
    node->sum += val * (r - l);
    node->add += val;
}

void push(Node* node, ll l, ll r) {
    if (node->add == 0) {
        return;
    }
    ll m = (l + r) / 2;
    if (!node->left) {
        node->left = new Node();
    }
    if (!node->right) {
        node->right = new Node();
    }
    apply(node->left, l, m, node->add);
    apply(node->right, m, r, node->add);
    node->add = 0;
}

ImplicitSegmentTree() {
}

ImplicitSegmentTree(ll N) {
    n = N;
}

ImplicitSegmentTree(const vector<ll>& start) {
    n = start.size();
    for (ll i = 0; i < n; i++) {
        change(i, start[i]);
    }
}

```

```

}

void change(Node*& node, ll l, ll r, ll id, ll val) {
    if (!node) {
        node = new Node();
    }
    if (l + 1 == r) {
        node->sum = val;
        node->add = 0;
        return;
    }
    push(node, l, r);
    ll m = (l + r) / 2;
    if (id < m) {
        change(node->left, l, m, id, val);
    } else {
        change(node->right, m, r, id, val);
    }
    fix(node);
}

void change(ll id, ll val) {
    change(root, 0, n, id, val);
}

void add(Node*& node, ll l, ll r, ll ql, ll qr, ll val) {
    if (qr <= l || r <= ql) {
        return;
    }
    if (!node) {
        node = new Node();
    }
    if (ql <= l && r <= qr) {
        apply(node, l, r, val);
        return;
    }
    push(node, l, r);
    ll m = (l + r) / 2;
    add(node->left, l, m, ql, qr, val);
    add(node->right, m, r, ql, qr, val);
    fix(node);
}

```

```

void add(ll ql, ll qr, ll val) {
    add(root, 0, n, ql, qr, val);
}

Node query(Node* node, ll l, ll r, ll ql, ll qr) {
    if (!node || qr <= l || r <= ql) {
        return neutral;
    }
    if (ql <= l && r <= qr) {
        return *node;
    }
    push(node, l, r);
    ll m = (l + r) / 2;
    return merge(query(node->left, l, m, ql, qr), query(node->right,
m, r, ql, qr));
}

Node query(ll ql, ll qr) {
    return query(root, 0, n, ql, qr);
}
};

```

5.8 Persistent DSU

```

struct PersistentDSU {
    ll n;
    vector<ll> prev, size;

    PersistentDSU(ll N) : n(N) {
        prev.assign(n, -1);
        size.assign(n, 1);
    }

    ll root(ll a) {
        while (prev[a] != -1)
            a = prev[a];
        return a;
    }

    bool connected(ll a, ll b) {
        return root(a) == root(b);
    }

    vector<pair<ll, ll>> changes; //

```

```

    bool unite(ll a, ll b) {
        a = root(a);
        b = root(b);
        if (a == b)
            return false;
        if (size[a] < size[b])
            swap(a, b);
        prev[b] = a;
        size[a] += size[b];
        changes.push_back({b, a});
        return true;
    }

    bool rollback() {
        if (changes.empty())
            return false;
        pair<ll, ll> ba = changes.back();
        changes.pop_back();
        prev[ba.first] = -1;
        size[ba.second] -= size[ba.first];
        return true;
    }
};

```

5.9 Prefix Automate

```

struct PrefixAutomate {
    const static int A = 26;

    struct Node {
        int next[A] = {-1};
        int link = -1;

        int prev = -1;
        int pe = -1;

        int has = 0;
    };

    vector<Node> tree;

    PrefixAutomate() {
        tree.push_back(Node());
    }

```

```

}

void add(const string& s) {
    int v = 0;
    for (char c : s) {
        int e = c - 'a';
        if (tree[v].next[e] == -1) {
            tree[v].next[e] = tree.size();
            tree.push_back(Node());
            tree.back().prev = v;
            tree.back().pe = e;
        }
        v = tree[v].next[e];
    }
    tree[v].has = 1;
}

void addVect(const vector<string>& vs) {
    for (const auto& s : vs) {
        add(s);
    }
}

void del(const string& s) {
    int v = 0;
    for (char c : s) {
        int e = c - 'a';
        if (tree[v].next[e] == -1) {
            return;
        }
        v = tree[v].next[e];
    }
    tree[v].has = 0;
}

void ahoCorasick() {
    queue<int> que;

    tree[0].prev = 0;
    tree[0].pe = -1;

    tree[0].link = 0;
    for (int i = 0; i < A; ++i) {
        if (tree[0].next[i] == -1) {

```

```

            tree[0].next[i] = 0;
        } else {
            que.push(tree[0].next[i]);
        }
    }

    while (!que.empty()) {
        int v = que.front();
        que.pop();

        if (tree[v].link == -1) {
            if (v == 0 || tree[v].prev == 0)
                tree[v].link = 0;
            else
                tree[v].link = tree[tree[v].prev].next[tree[v].pe];
        }

        for (int i = 0; i < A; ++i) {
            if (tree[v].next[i] == -1) {
                tree[v].next[i] = tree[tree[v].link].next[i];
            } else {
                que.push(tree[v].next[i]);
            }
        }
    }
};

```

5.10 Segment Tree Beats

```

struct SegmentTreeBeats {
    struct Node {
        ll val = 0;
        ll firstMax = -inf;
        ll cntMax = 0;
        ll secondMax = -inf;

        ll min_eq = inf;

        Node(ll Val = 0, ll FirstMax = -inf, ll CntMax = 0, ll SecondMax
= -inf) {
            val = Val;
            firstMax = FirstMax;
            cntMax = CntMax;

```

```

        secondMax = SecondMax;
    }
};

Node fromOne(ll a) {
    return Node(a, a, 1, -inf);
}

Node neutral = Node();
ll n;
vector<Node> tree;

SegmentTreeBeats(const vector<ll>& start) {
    n = start.size();
    tree.resize(4 * n + 3);
    init(start);
}

SegmentTreeBeats(ll N) {
    n = N;
    tree.resize(4 * n + 3);
    vector<ll> start(n, 0);
    init(start);
}

Node merge(Node n1, Node n2) {
    Node res(n1.val + n2.val);

    if (n1.firstMax > n2.firstMax) {
        res.firstMax = n1.firstMax;
        res.cntMax = n1.cntMax;
        res.secondMax = max(n1.secondMax, n2.firstMax);
    } else if (n1.firstMax < n2.firstMax) {
        res.firstMax = n2.firstMax;
        res.cntMax = n2.cntMax;
        res.secondMax = max(n1.firstMax, n2.secondMax);
    } else {
        res.firstMax = n1.firstMax;
        res.cntMax = n1.cntMax + n2.cntMax;
        res.secondMax = max(n1.secondMax, n2.secondMax);
    }
    return res;
}

```

```

void fix(ll v, ll l, ll r) {
    tree[v] = merge(tree[v * 2 + 1], tree[v * 2 + 2]);
}

void init(ll v, ll l, ll r, const vector<ll>& start) {
    if (l + 1 == r) {
        tree[v] = fromOne(start[l]);
        return;
    }

    ll m = (r + 1) / 2;
    init(v * 2 + 1, l, m, start);
    init(v * 2 + 2, m, r, start);
    fix(v, l, r);
}

void init(const vector<ll>& start) {
    init(0, 0, n, start);
}

// [l: r)
Node query(ll v, ll l, ll r, ll ql, ll qr) {
    if (ql <= l && r <= qr) {
        return tree[v];
    }
    if (r <= ql || qr <= l) {
        return neutral;
    }

    ll m = (r + 1) / 2;
    push(v, l, r);
    return merge(query(v * 2 + 1, l, m, ql, qr), query(v * 2 + 2, m,
r, ql, qr));
}

Node query(ll ql, ll qr) {
    return query(0, 0, n, ql, qr);
}

bool breakCondition(ll v, ll l, ll r, ll val) {
    return tree[v].firstMax <= val;
}

bool tagCondition(ll v, ll l, ll r, ll val) {

```

```

    return tree[v].secondMax < val;
}

void apply(ll v, ll l, ll r, ll val) {
    // only if tagCondition() == true
    ll diff = min(tree[v].firstMax, val) - tree[v].firstMax;
    tree[v].val += tree[v].cntMax * diff;
    tree[v].firstMax += diff;

    tree[v].min_eq = min(val, tree[v].min_eq);
}

void push(ll v, ll l, ll r) {
    ll m = (r + 1) / 2;

    apply(v * 2 + 1, l, m, tree[v].min_eq);
    apply(v * 2 + 2, m, r, tree[v].min_eq);
    tree[v].min_eq = inf;
}

void update(ll v, ll l, ll r, ll ql, ll qr, ll val) {
    if (r <= ql || qr <= l || breakCondition(v, l, r, val)) {
        return;
    }
    if (ql <= l && r <= qr && tagCondition(v, l, r, val)) {
        apply(v, l, r, val);
        return;
    }

    ll m = (r + 1) / 2;
    push(v, l, r);
    update(v * 2 + 1, l, m, ql, qr, val);
    update(v * 2 + 2, m, r, ql, qr, val);
    fix(v, l, r);
}

void update(ll ql, ll qr, ll val) {
    update(0, 0, n, ql, qr, val);
}

pair<ll, ll> descent(ll v, ll l, ll r, ll ql, ll qr, ll x) {
    if (r <= ql || qr <= l)
        return {-1, x};
    if (ql <= l && r <= qr) {

```

```

        if (tree[v].val < x)
            return {-1, x - tree[v].val};
        if (l + 1 == r)
            return {l + 1, x};
    }
    if (l + 1 == r)
        return {-1, x};

    ll m = (r + 1) / 2;
    push(v, l, r);
    auto left = descent(v * 2 + 1, l, m, ql, qr, x);
    if (left.first != -1)
        return left;
    return descent(v * 2 + 2, m, r, ql, qr, left.second);
}

ll descent(ll ql, ll qr, ll x) {
    if (x <= 0)
        return ql;
    return descent(0, 0, n, ql, qr, x).first;
}
};

```

5.11 Segment Tree Mass

```

struct SegmentTreeMass {
    struct Node {
        ll val = inf;
        ll add = 0;

        Node(ll Val = inf, ll Add = 0) {
            val = Val;
            add = Add;
        }
    };

    Node neutral = Node(inf, 0);
    ll n;
    vector<Node> tree;

    SegmentTreeMass(const vector<ll>& start) {
        n = start.size();
        tree.resize(4 * n + 3);
        init(start);
    }
};

```

```

}

SegmentTreeMass(11 N) {
    n = N;
    tree.resize(4 * n + 3);
    vector<11> start(n, 0);
    init(start);
}

Node merge(Node n1, Node n2) {
    return Node(min(n1.val, n2.val));
}

void fix(11 v, 11 l, 11 r) {
    tree[v] = merge(tree[v * 2 + 1], tree[v * 2 + 2]);
}

void apply(11 v, 11 l, 11 r, 11 val) {
    tree[v].add += val;
    tree[v].val += val;
}

void push(11 v, 11 l, 11 r) {
    11 m = (r + 1) / 2;

    apply(v * 2 + 1, l, m, tree[v].add);
    apply(v * 2 + 2, m, r, tree[v].add);
    tree[v].add = 0;
}

void init(11 v, 11 l, 11 r, const vector<11>& start) {
    if (l + 1 == r) {
        tree[v] = Node(start[l], 0);
        return;
    }

    11 m = (r + 1) / 2;
    init(v * 2 + 1, l, m, start);
    init(v * 2 + 2, m, r, start);
    fix(v, l, r);
}

void init(const vector<11>& start) {
    init(0, 0, n, start);
}

}

// [l: r)
Node query(11 v, 11 l, 11 r, 11 ql, 11 qr) {
    if (ql <= l && r <= qr) {
        return tree[v];
    }
    if (r <= ql || qr <= l) {
        return neutral;
    }

    11 m = (r + 1) / 2;
    push(v, l, r);
    return merge(query(v * 2 + 1, l, m, ql, qr), query(v * 2 + 2, m,
r, ql, qr));
}

Node query(11 ql, 11 qr) {
    return query(0, 0, n, ql, qr);
}

void add(11 v, 11 l, 11 r, 11 ql, 11 qr, 11 val) {
    if (ql <= l && r <= qr) {
        apply(v, l, r, val);
        return;
    }
    if (r <= ql || qr <= l) {
        return;
    }

    11 m = (r + 1) / 2;
    push(v, l, r);
    add(v * 2 + 1, l, m, ql, qr, val);
    add(v * 2 + 2, m, r, ql, qr, val);
    fix(v, l, r);
}

void add(11 ql, 11 qr, 11 val) {
    add(0, 0, n, ql, qr, val);
}

11 descent(11 v, 11 l, 11 r, 11 ql, 11 qr, 11 x) {
    if (r <= ql || qr <= l)
        return -1;
}

```

```

    if (tree[v].val >= x)
        return -1;
    if (l + 1 == r)
        return 1;

    ll m = (r + 1) / 2;
    push(v, l, r);
    ll left = descent(v * 2 + 1, l, m, ql, qr, x);
    if (left != -1)
        return left;
    return descent(v * 2 + 2, m, r, ql, qr, x);
}

ll descent(ll ql, ll qr, ll x) {
    return descent(0, 0, n, ql, qr, x);
}
};

```

5.12 Segment Tree

```

struct SegmentTree {
    struct Node {
        ll val;

        Node(ll Val = inf) : val(Val) {}
    };

    Node neutral = Node();

    ll n;
    vector<Node> tree;

    void init(ll v, ll l, ll r, const vector<ll>& start) {
        if (l + 1 == r) {
            tree[v] = Node(start[l]);
            return;
        }

        ll m = (l + r) / 2;

        init(v * 2 + 1, l, m, start);
        init(v * 2 + 2, m, r, start);
        tree[v] = merge(tree[v * 2 + 1], tree[v * 2 + 2]);
    }
};

```

```

}

SegmentTree() {}

SegmentTree(const vector<ll>& start) {
    n = start.size();
    tree.resize(n * 4 + 3);
    init(0, 0, n, start);
}

Node merge(Node n1, Node n2) {
    return Node(min(n1.val, n2.val));
}

Node query(ll v, ll l, ll r, ll ql, ll qr) {
    if (ql <= l && r <= qr)
        return tree[v];
    if (qr <= l || r <= ql)
        return neutral;

    ll m = (l + r) / 2;
    return merge(query(v * 2 + 1, l, m, ql, qr), query(v * 2 + 2, m,
r, ql, qr));
}

Node query(ll ql, ll qr) {
    return query(0, 0, n, ql, qr);
}

void change(ll v, ll l, ll r, ll id, ll val) {
    if (l + 1 == r && l == id) {
        tree[v] = Node(val);
        return;
    }
    if (r <= id || id < l) {
        return;
    }

    ll m = (l + r) / 2;
    change(v * 2 + 1, l, m, id, val);
    change(v * 2 + 2, m, r, id, val);
    tree[v] = merge(tree[v * 2 + 1], tree[v * 2 + 2]);
}

```

```

void change(ll id, ll val) {
    change(0, 0, n, id, val);
}

ll descent(ll v, ll l, ll r, ll ql, ll qr, ll x) {
    if (r <= ql || qr <= l)
        return -1;
    if (tree[v].val >= x)
        return -1;
    if (l + 1 == r)
        return 1;

    ll m = (l + r) / 2;
    ll left = descent(v * 2 + 1, l, m, ql, qr, x);
    if (left != -1)
        return left;
    return descent(v * 2 + 2, m, r, ql, qr, x);
}

ll descent(ll ql, ll qr, ll x) {
    return descent(0, 0, n, ql, qr, x);
}
};

```

5.13 Sparse Table

```

template <class T>
struct SparseTable {
    ll n, k;
    vector<T> a;
    vector<vector<T>> sparse;
    vector<ll> length;

    SparseTable(vector<T>& start) {
        n = start.size();
        k = 0;
        for (ll d = 1; d <= n; d *= 2, ++k) {}
        a.assign(start.begin(), start.end());
        sparse.assign(n, vector<T>(k));
        countSparse();
        precalc();
    }
};

```

```

void precalc() {
    length.assign(n + 1, 0);
    ll t = 0;
    for (int l = 1; l <= n; ++l) {
        if ((1ll << (t + 1)) < l)
            ++t;
        length[l] = t;
    }
}

void countSparse() {
    for (int i = 0; i < n; ++i) {
        sparse[i][0] = a[i];
    }
    for (int j = 1; j < k; ++j) {
        for (int i = 0; i < n; ++i) {
            ll d = (1ll << (j - 1));
            sparse[i][j] = max(sparse[i][j - 1], sparse[min(n - 1, i
+ d)][j - 1]);
        }
    }
}

T query(ll l, ll r) {
    ll t = length[r - 1];

    ll d = (1ll << t);

    return max(sparse[l][t], sparse[r - d][t]);
}
};

```

5.14 Treap

```

random_device rd;
mt19937 mersenne(rd());

struct Treap {
    Treap* left;
    Treap* right;
    ll x;
    unsigned int y;
    unsigned int size;
};

```

```

11 dp;

Treap(11 val = 0) {
    left = nullptr;
    right = nullptr;
    size = 1;
    x = val;
    y = mersenne();
    update(this);
}

friend 11 get_dp(Treap* node) {
    if (node == nullptr)
        return 0;
    return node->dp;
}

friend unsigned int get_size(Treap* node) {
    if (node == nullptr)
        return 0;
    return node->size;
}

friend void update(Treap* node) {
    if (node == nullptr)
        return;
    node->size = 1 + get_size(node->left) + get_size(node->right);
    node->dp = max({node->x, get_dp(node->left), get_dp(node->right)});
};

// split by size
friend pair<Treap*, Treap*> split_k(Treap* node, unsigned int k) {
    if (node == nullptr)
        return {nullptr, nullptr};
    if (get_size(node->left) < k) {
        auto res = split_k(node->right, k - get_size(node->left) - 1);
        node->right = res.first;
        update(node);
        return {node, res.second};
    } else {
        auto res = split_k(node->left, k);
        node->left = res.second;
        update(node);
        return {res.first, node};
    }
}

```

```

        update(node);
        return {res.first, node};
    }
}

// split by dp: dp in left < k
friend pair<Treap*, Treap*> split_dp(Treap* node, unsigned int k) {
    if (node == nullptr)
        return {nullptr, nullptr};
    if (get_dp(node) < k) {
        auto res = split_dp(node->right, k);
        node->right = res.first;
        update(node);
        return {node, res.second};
    } else {
        auto res = split_dp(node->left, k);
        node->left = res.second;
        update(node);
        return {res.first, node};
    }
}

friend Treap* merge(Treap* left, Treap* right) {
    if (left == nullptr)
        return right;
    if (right == nullptr)
        return left;
    if (left->y > right->y) {
        left->right = merge(left->right, right);
        update(left);
        return left;
    } else {
        right->left = merge(left, right->left);
        update(right);
        return right;
    }
}

friend Treap* insert(Treap* node, unsigned int pos, 11 val) {
    auto res = split_k(node, pos);
    Treap* tr = new Treap(val);
    return merge(merge(res.first, tr), res.second);
}

```

```

friend Treap* erase(Treap* node, unsigned int pos) {
    auto res = split_k(node, pos + 1);
    auto left_res = split_k(res.first, pos);
    return merge(left_res.first, res.second);
}

// return {element, root}
friend pair<Treap*, Treap*> kth_element(Treap* node, unsigned int k)
{
    if (get_size(node) < k)
        return {nullptr, node};
    auto res = split_k(node, k + 1);
    auto left_res = split_k(res.first, k);
    return {left_res.second, merge(left_res.first, merge(left_res.
second, res.second))};
}

// return sub segment [l, r) and remaineng Treap [0, l) + [r + 1,
size)
friend pair<Treap*, Treap*> cut(Treap* node, unsigned int l, unsigned
int r) {
    auto res = split_k(node, r);
    auto left_res = split_k(res.first, l);
    return {left_res.second, merge(left_res.first, res.second)};
}
};

```

6 Flows

6.1 Dinic

```

struct FlowSolver {
    struct DinicEdge {
        ll a, b, cap, flow;
    };

    vector<DinicEdge> edges;
    vector<vector<ll>> g;
    ll n;

    FlowSolver(ll N) {
        n = N;
        g.resize(n);
    }
};

```

```

}

void add_edge(ll u, ll v, ll w) {
    g[u].push_back(edges.size());
    edges.push_back({u, v, w, 0});
    g[v].push_back(edges.size());
    edges.push_back({v, u, 0, 0});
}

bool bfs(ll s, ll t, vector<ll>& d) {
    queue<ll> q;
    d.assign(n, -1);
    d[s] = 0;
    q.push(s);

    while (!q.empty()) {
        ll v = q.front();
        q.pop();

        for (ll id : g[v]) {
            ll u = edges[id].b;
            if (d[u] == -1) {
                if (edges[id].cap > edges[id].flow) {
                    d[u] = d[v] + 1;
                    q.push(u);
                }
            }
        }
    }

    return d[t] != -1;
}

ll dfs(ll v, ll t, ll flow, vector<ll>& d, vector<ll>& ptr) {
    if (flow == 0) {
        return 0;
    }
    if (v == t) {
        return flow;
    }

    for (; ptr[v] < g[v].size(); ++ptr[v]) {
        ll id = g[v][ptr[v]];
        ll u = edges[id].b;
    }
}

```

```

        if (d[u] == d[v] + 1) {
            ll add_flow = dfs(u, t, min(flow, edges[id].cap - edges[
id].flow), d, ptr);
            if (add_flow > 0) {
                edges[id].flow += add_flow;
                edges[id ^ 1].flow -= add_flow;
                return add_flow;
            }
        }
    }
    return 0;
}

ll dinic(ll s, ll t) {
    ll flow = 0;
    vector<ll> d;
    while (bfs(s, t, d)) {
        vector<ll> ptr(n, 0);
        while (ll add_flow = dfs(s, t, inf, d, ptr)) {
            flow += add_flow;
        }
    }
    return flow;
}
};

```

7 Graphs

7.1 2 SAT

```

struct SATSolver {
    ll n;
    vector<vector<ll>> g, gt;
    vector<char> used;
    vector<ll> order, comp;

    void init(ll N) {
        n = N;
        g.assign(2 * n, {});
        gt.assign(2 * n, {});
    }
}

```

```

// literal: var in [0..n), val=false -> 2*var, val=true -> 2*var+1
ll lit(ll var, ll val) {
    return 2 * var + (val ? 1 : 0);
}

ll neg(ll x) {
    return x ^ 1;
}

void add_imp(ll a, ll b) {
    g[a].push_back(b);
    gt[b].push_back(a);
}

// (a_var is a_val) OR (b_var is b_val)
void add_or(ll a_var, ll a_val, ll b_var, ll b_val) {
    ll a = lit(a_var, a_val);
    ll b = lit(b_var, b_val);
    add_imp(neg(a), b);
    add_imp(neg(b), a);
}

void dfs1(ll v) {
    used[v] = 1;
    for (ll to : g[v])
        if (!used[to])
            dfs1(to);
    order.push_back(v);
}

void dfs2(ll v, ll cl) {
    comp[v] = cl;
    for (ll to : gt[v])
        if (comp[to] == -1)
            dfs2(to, cl);
}

// returns empty vector if unsat; otherwise assignment[i] in {0,1}
vector<ll> solve() {
    used.assign(2 * n, 0);
    order.clear();
    for (ll i = 0; i < 2 * n; ++i)
        if (!used[i])
            dfs1(i);
}

```

```

    comp.assign(2 * n, -1);
    ll j = 0;
    for (ll i = 2 * n - 1; i >= 0; --i) {
        ll v = order[i];
        if (comp[v] == -1)
            dfs2(v, j++);
    }

    vector<ll> ans(n);
    for (ll i = 0; i < n; ++i) {
        if (comp[2 * i] == comp[2 * i + 1])
            return {};
        ans[i] = (comp[2 * i] < comp[2 * i + 1]);
    }
    return ans;
}
};

```

7.2 Euler Path

```

void cycle(ll prev, ll v, vector<vector<pair<ll, ll>>>& g, vector<ll>&
ans) {
    while (!g[v].empty()) {
        auto iu = g[v].back();
        ll u = iu.second;
        g[v].pop_back();
        cycle(iu.first, u, g, ans);
    }
    if (prev >= 0)
        ans.push_back(prev);
}

```

7.3 Hopcroft Karp

```

struct HopcroftKarp {
    ll n1, n2;
    vector<vector<ll>> g;
    vector<ll> dist;
    vector<ll> mt1, mt2;

    void init(ll N1, ll N2) {
        n1 = N1;
        n2 = N2;
    }
}

```

```

    g.assign((int)n1, {});
    mt1.assign((int)n1, -1);
    mt2.assign((int)n2, -1);
    dist.assign((int)n1, 0);
}

void add_edge(ll u, ll v) {
    // u in [0..n1), v in [0..n2)
    g[(int)u].push_back(v);
}

bool bfs() {
    queue<ll> q;
    for (ll i = 0; i < n1; ++i) {
        if (mt1[(int)i] == -1) {
            dist[(int)i] = 0;
            q.push(i);
        } else {
            dist[(int)i] = -1;
        }
    }

    bool found = false;
    while (!q.empty()) {
        ll v = q.front();
        q.pop();

        for (ll to : g[(int)v]) {
            ll u = mt2[(int)to];
            if (u == -1) {
                found = true;
            } else if (dist[(int)u] == -1) {
                dist[(int)u] = dist[(int)v] + 1;
                q.push(u);
            }
        }
    }

    return found;
}

bool dfs(ll v) {
    for (ll to : g[(int)v]) {
        ll u = mt2[(int)to];
        if (u == -1 || (dist[(int)u] == dist[(int)v] + 1 && dfs(u)))

```

```

{
    mt1[(int)v] = to;
    mt2[(int)to] = v;
    return true;
}
}
dist[(int)v] = -1;
return false;
}

ll solve() {
    ll matching = 0;
    while (bfs()) {
        for (ll i = 0; i < n1; ++i) {
            if (mt1[(int)i] == -1) {
                if (dfs(i)) {
                    ++matching;
                }
            }
        }
    }
    return matching;
}
};

// run step of dfs from i0
for (int j = 1; j <= m; ++j) {
    if (used[j]) {
        continue;
    }

    int cur = a[i0][j] - u[i0] - v[j];
    if (cur < minv[j]) {
        minv[j] = cur;
        way[j] = j0;
    }
    if (minv[j] < delta) {
        delta = minv[j];
        j1 = j;
    }
}

// update potentials
for (int j = 0; j <= m; ++j) {
    if (used[j]) {
        u[p[j]] += delta;
        v[j] -= delta;
    } else {
        minv[j] -= delta;
    }
}

j0 = j1;
} while (p[j0] != 0);

while (j0) {
    int j1 = way[j0];
    p[j0] = p[j1];
    j0 = j1;
}

return {-v[0], p};
}

```

7.4 Hungarian

```

// a size is (n + 1) x (m + 1), 1-indexing
// n <= m; if n < m than a[i][j] must be >= 0
pair<ll, vector<int>> hungarian(const vector<vector<ll>>& a, int n, int m) {
    vector<ll> u(n + 1, 0), v(m + 1, 0);
    vector<int> p(m + 1, 0), way(m + 1, 0);
    for (int i = 1; i <= n; ++i) {
        int j0 = 0;
        p[j0] = i;

        vector<char> used(m + 1, 0);
        vector<ll> minv(m + 1, inf);
        do {
            used[j0] = true;
            int i0 = p[j0];
            int j1 = -1, delta = inf;

```

7.5 Initial Pairing

```

int init_pairing(int n1, int n2, const vector<vector<int>>& g, vector<int>
    >& mt, vector<int>& rmt) {

```

```

auto PACK = [](int x, int y) { return ((1ll)x << 30) + y; };
auto SUNPACK = [](int x) {
    const static ll mask = (1ll << 30) - 1;
    return x & mask;
};

vector<int> deg(n1, 0);
vector<vector<int>> ng(n1);
vector<vector<pair<int, int>>> rg(n2);
vector<int> next(n1, 0);
set<ll> que;
for (int i = 0; i < n1; ++i) {
    for (auto j : g[i]) {
        ++deg[i];
        rg[j].push_back({i, ng[i].size()});
        ng[i].push_back(j);
    }
    que.insert(PACK(deg[i], i));
}

int pairing = 0;

while (!que.empty()) {
    auto res = *que.begin();
    que.erase(res);

    int i = SUNPACK(res);
    if (deg[i] == 0) {
        continue;
    }

    while (ng[i][next[i]] == -1) {
        ++next[i];
    }
    int j = ng[i][next[i]++];

    deg[i] = 0;
    for (auto ks : rg[j]) {
        int k = ks.first;
        int sz = ks.second;
        if (deg[k] != 0) {
            ng[k][sz] = -1;
            que.erase(PACK(deg[k], k));
            --deg[k];

```

```

        que.insert(PACK(deg[k], k));
    }
}
mt[j] = i;
rmt[i] = j;
++pairing;
}
return pairing;
}

vector<int> fast_kuhn(int n1, int n2, vector<vector<int>>& g) {
    vector<char> used(n1, 0);
    vector<int> mt(n2, -1), rmt(n1, -1);

    init_pairing(n1, n2, g, mt, rmt);

    int mark = 0;

    for (int run = 1; run;) {
        ++mark;
        run = 0;
        for (int i = 0; i < n1; ++i) {
            if (rmt[i] == -1 && kuhn_dfs(i, used, mark, g, mt)) {
                run = 1;
            }
        }

        for (int j = 0; j < n2; ++j) {
            if (mt[j] != -1) {
                rmt[mt[j]] = j;
            }
        }

        return mt;
    }
}

```

7.6 Kuhn

```

bool kuhn_dfs(int v, vector<int>& used, int mark, vector<vector<int>>& g,
    vector<int>& mt) {
    if (used[v] == mark) {
        return false;
    }

```

```

used[v] = mark;

for (int u : g[v]) {
    if (mt[u] == -1 || kuhn_dfs(mt[u], used, mark, g, mt)) {
        mt[u] = v;
        return true;
    }
}
return false;
}

vector<int> kuhn(int n1, int n2, vector<vector<int>>& g) {
    vector<int> used(n1, 0);
    vector<int> mt(n2, -1);

    int mark = 1;

    for (int i = 0; i < n1; ++i) {
        if (kuhn_dfs(i, used, mark, g, mt)) {
            ++mark;
        }
    }

    return mt;
}

```

8 Number Theory

8.1 Eratosthenes

```

void erat(ll N, vector<ll>& primes, vector<ll>& mi) {
    mi.assign(N, 0);
    for (int i = 2; i < N; ++i) {
        if (mi[i] == 0) {
            primes.push_back(i);
            mi[i] = i;
        }

        for (int j = 0; j < primes.size() && primes[j] <= mi[i] && i *
primes[j] < N; ++j) {
            mi[primes[j] * i] = primes[j];
        }
    }
}

```

```

}

```

8.2 FloorSum

```

ll floor_div(ll a, ll b) { // b > 0
    if (a >= 0)
        return a / b;
    return (a - b + 1) / b;
}

// sum_{x=0}^{n-1} floor((a*x + b) / m), m > 0
ll floor_sum(ll n, ll m, ll a, ll b) {
    if (n <= 0) return 0;
    ll ans = 0;
    auto pull = [&](ll &x, ll w) {
        ll q = floor_div(x, m);
        ans += q * w;
        x -= q * m;
    };
    while (true) {
        pull(a, n * (n - 1) / 2);
        pull(b, n);
        ll y = a * n + b;
        if (y < m) return ans;
        n = y / m;
        b = y % m;
        swap(a, m);
    }
}

```

8.3 Miller Rabin

```

bool _miller_rabin_test(ll n, ll d, ll s, ll a) {
    ll x = bin_pow_mod(a, d, n);
    for (int i = 0; i < s; ++i) {
        ll y = ((__int128_t)x * x) % n;
        if (y == 1 && x != 1) {
            return false;
        }

        x = y;
    }
    if (x != 1) {

```

```

        return false;
    }
    return true;
}

bool is_prime_ml(ll n) {
    if (n <= 1) {
        return false;
    }
    // all primes
    vector<ll> as = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37};
    ll d = n - 1;
    ll s = 0;
    while ((d & 1) == 0) {
        ++s;
        d >>= 1;
    }
    for (ll a : as) {
        if (n == a) {
            return true;
        }
        if (n % a == 0) {
            return false;
        }
        if (!_miller_rabin_test(n, d, s, a)) {
            return false;
        }
    }
    return true;
}

```

9 Strings

9.1 LCP

```

// p - suffix massive output
// c - suffix massive classes
vector<ll> calc_lcp(ll n, const string& s, const vector<ll>& c, const
    vector<ll>& p) {
    vector<ll> lcp(n - 1, 0);

    ll pref = 0;

```

```

    for (int i = 0; i < n - 1; ++i) {
        ll pos = c[i];
        ll j = p[pos - 1];
        while (i + pref < n && j + pref < n && s[i + pref] == s[j + pref]) {
            ++pref;
        }
        lcp[pos - 1] = pref;
        pref = max(0ll, pref - 1);
    }

    return lcp;
}

```

9.2 Prefix Function

```

vector<ll> prefix_function(string s) {
    ll n = s.size();
    vector<ll> p(n, 0);
    for (int i = 1; i < n; ++i) {
        ll k = p[i - 1];
        while (k > 0 && s[i] != s[k])
            k = p[k - 1];
        if (s[i] == s[k])
            ++k;
        p[i] = k;
    }
    return p;
}

```

9.3 Suffix Massive

```

void radix_sort(vector<pair<pair<ll, ll>, ll>>& prs) {
    ll n = prs.size();
    vector<ll> cnt(n, 0);
    for (auto mem : prs) {
        cnt[mem.first.first]++;
    }
    vector<ll> ptr(n, 0);
    for (int i = 1; i < n; ++i) {
        ptr[i] = ptr[i - 1] + cnt[i - 1];
    }
}

```

```

vector<pair<pair<ll, ll>, ll>> new_prs(n);

for (auto mem : prs) {
    ll i = mem.first.first;

    new_prs[ptr[i]++] = mem;
}
prs = new_prs;
}

vector<ll> suffix_array(string s) {
    s += '$';

    ll n = s.size();

    vector<ll> p(n), c(n);

    { // k == 0
        vector<pair<ll, ll>> prs(n);
        for (int i = 0; i < n; ++i) {
            prs[i] = {s[i], i};
        }
        sort(all(prs));
        for (int i = 0; i < n; ++i) {
            p[i] = prs[i].second;
        }

        c[p[0]] = 0;
        for (int i = 1; i < n; ++i) {
            if (prs[i].first == prs[i - 1].first) {
                c[p[i]] = c[p[i - 1]];
            } else {
                c[p[i]] = c[p[i - 1]] + 1;
            }
        }
    }

    ll k = 0;

    while ((1ll << k) < n) {
        vector<pair<pair<ll, ll>, ll>> prs(n);
        for (int i = 0; i < n; ++i) {
            ll t = p[i];
            ll j = ((t - (1ll << k)) % n + n) % n;
            prs[i] = {{c[j], c[t]}, j};

```

```

        }
        radix_sort(prs);

        for (int i = 0; i < n; ++i) {
            p[i] = prs[i].second;
        }

        c[p[0]] = 0;
        for (int i = 1; i < n; ++i) {
            if (prs[i].first == prs[i - 1].first) {
                c[p[i]] = c[p[i - 1]];
            } else {
                c[p[i]] = c[p[i - 1]] + 1;
            }
        }

        ++k;
    }

    vector<ll> ans;
    ans.reserve((int)n - 1);
    for (int i = 1; i < n; ++i)
        ans.push_back(p[i]);
    return ans;
}

9.4 Z Function

vector<ll> z_function(string s) {
    ll n = s.size();
    vector<ll> z(n, 0);
    ll r = 0;
    for (int i = 1; i < n; ++i) {
        if (z[i - r] < r + z[r] - i)
            z[i] = z[i - r];
        else {
            z[i] = max(r + z[r] - i, 0ll);
            while (i + z[i] < n && s[i + z[i]] == s[z[i]])
                ++z[i];
            r = i;
        }
    }
    return z;
}

```